Technical Project: Development of a New Musical Wind Controller Using DIY and 'Maker' Frameworks

STS Project: The Technological Momentum of Nyle Steiner's Electronic Valve Instrument (EVI)

A Thesis Prospectus submitted to the Department of Engineering and Society

Presented to the Faculty of the School of Engineering and Applied Science

University of Virginia: Charlottesville, Virginia

In Partial Fulfillment of the Requirements for the Degree

Bachelor of Science in Mechanical Engineering

John Anthony (Bob) Kammauff

Summer, 2026

On my honor as a University student, I have neither given nor received unauthorized aid on this assignment as defined by the Honor Guidelines for Thesis-Related Assignments.

Advisors

Ben Laugelli, Department of Engineering and Society

Introduction

In the modern musical climate, a family of instruments known as wind controllers[1] serve as the bridge between wind musicians and electronic music by electrically mapping the nuances of breath control to musical expression. Unfortunately, the market is extremely lacking in models for students who value affordability and understanding, and without student development, the instrument risks falling further into obscurity. In recent years, many academic institutions have spearheaded STEM project-based learning programs, which, when combined with the 'Maker' movement, have made basic prototyping skills and embedded electronic system knowledge commonplace.

I am proposing the development of wind controller designs geared towards this community of educators, students, and hobbyists. This will take the form of a microcontroller project that could be easily recreated by a user familiar with soldering and 3D printing to grow the userbase of new wind controller players. The goal is to create a body of design documentation that can continue to be iterated upon to create better educational and commercial wind controller products for future generations.

Since this technical project requires revisiting core design considerations, a historical body of knowledge surrounding the development of wind controllers is needed to ensure compatibility and continuity across generations. This will involve collecting sources surrounding the invention of the first wind controller by Nyle Steiner and analyzing the information through the STS framework Technical Momentum.

---

[1] For the sake of clarity, wind controller will be used regardless of if the device has an onboard synthesizer (where 'wind synth' would be the more appropriate term).

Because the challenge of developing a musical instrument is sociotechnical in nature, it requires attending to both its technical and social aspects to accomplish successfully. In what follows, I set out two related research proposals: a technical project proposal for developing an accessible, yet robust modular wind controller design with off-the-shelf components and, second, a STS project proposal analyzing the lasting influences from the development of the first wind controller, the electronic valve instrument.

Technical Project Proposal

The basic concept behind a wind controller takes familiar wind instrument bodies and replaces the keys and mouthpiece with transducers to generate or control electronic sound. These instruments allow wind musicians the enhanced tonal capabilities of electronic sound in a controller familiar to them. Most wind controllers are modeled off woodwind instruments, such as the Akai EWI series (EWI Series, 2026), while the brass-style EVI only has one manufacturer: Berglund Instruments, custom-made in Sweden and sold by Patchman Music in the states (Traum M. , The Wind Controller FAQ, 2026). While these instruments are very well crafted, and the Akai models have the advantage of being integrated with their extensive sound library, they are not very affordable, with the Akai EWI Solo and 5000 model retailing at $549.99 and $899.99 (Sweetwater.com, 2026), and the NuEVI models from Berglund start at $1545 (Berglund Instruments, 2026). While these prices are comparable to other electronic instruments, they create a high barrier to entry for new users, especially those looking for an EVI. Users have tried developing DIY controllers, but most lack educational value regarding their findings, and make replication difficult. There is a wealth of academic research available to the public regarding sound synthesis algorithms for compelling electronic timbres (the Center for Computer Research in Music and Acoustics at Stanford is a great resource), however, that research "has not often connected back to how the musician will interact with [those algorithms]" (Taki, 2020). Without academic/open-source instrument design research, the wind controller risks remaining in this obscure monopoly.

The aim of this technical project is to reverse this trajectory by developing a collection of robust open-source wind controller designs using readily available electrical and mechanical components and accessible prototype manufacturing techniques (i.e. 3D printing). The multiplicity of designs will seek to capture the essential elements of different commercial wind controllers that can't be consolidated into a single design: multiple supported fingering styles (EWI vs. EVI), MIDI compatibility, and additional physical modulation/effect controls (bite sensors, slur keys, pitchbend, etc.). These designs will be fully configurable due to using low-cost microcontrollers with active user-developer bases, which will also serve to offload a lot of software design. The deliverables will consist of a repository containing at least 2 hardware designs for both a pure MIDI controller and controller with onboard synthesizer capabilities. The material costs for all designs will be under $100 to ensure accessibility.

This project will utilize concepts gathered from both the departments of Mechanical Engineering and Computer Composition Technologies in Music (CCT), as well as from personal research. The Electrosmith Daisy will be the main microcontroller in consideration for the main processor, as recommended by a CCT graduate student for its dedicated onboard DAC and low price (L. Hall, personal communication, May 6, 2026). The Daisy can be programmed using the online Daisy IDE using C++, and there is community support for running PureData patches by using the Heavy compiler (Electrosmith, 2024). PureData is an open-source graphical programming language which will help facilitate sound and GUI design, as well as making user development more accessible. A prototype EVI has already been developed using a Bela embedded system, which, while outside the price range for consideration, will serve as the first guiding iteration for this design project (Bob Kammauff, 2026). The finalized designs will be tested for latency and consistency with lab tests, with user trials being used to ensure that the design can be accurately recreated without active assistance from the development team. The musical viability of the designs will be showcased in both classical and popular contexts through a composition project taken on by myself.

STS Project Proposal

The wind controller, as we know it today, was invented by Nyle Steiner in 1975 (Traum M. , n.d.). While Steiner was actively recording projects well into the 2010's (Steiner N. , 2010), little of his work has been publicly documented. The only academic posting from Steiner is the abstract of a presentation that he gave at a conference in 2004 about his Electronic Valve Instrument (EVI) (Steiner N. A., 2004). While the EVI is commonly regarded as one of the best wind controllers, only one company is actively manufacturing this style: Berglund Instruments (Traum M. , The Wind Controller FAQ, 2026). Most documentation surrounding Steiner and his EVI comes from Matt Traum, author of 'The Nyle Steiner Homepage' hosted on his website patchmanmusic.com (Traum M. , The Nyle Steiner Homepage, 2025). While it is the most thorough documentation on Steiner, it connotes a heavy bias towards Steiner's designs. The following quote detailing Steiner's philosophy about control/fingering configurations summarizes how most of this website is written:

"*I recently spoke with Nyle Steiner, the father of the EWI and EVI, on this subject of fingering standardization and one of the most important things to him is that there is a standard fingering system / mechanism that is maintained on wind controllers. This is no small matter and should not be taken lightly. He likens it to a musician being able to sit down and play ANY MODEL piano or guitar and KNOWING that the fingering patterns they have learned on another model will translate to it as expected. So even though the Akai EWI4000s is not a true*

EVI with a canister, let us consider this fingering system very carefully in honor of Nyle's legacy that he has worked on for 35+ years and try to carry over all the fingerings from the EVI as would be expected and as best we can in the EWI form factor."

The above quote is also one of the few documented design decisions from both Traum and Steiner regarding their wind controllers. This is unfortunately a common case for electronic instrument hardware design, as noted by Daniel Taki: There is a lot of publicly available research surrounding sound synthesis algorithms, but most of the research surrounding the physical instrument design is proprietary (Taki, 2020). While the talent of Steiner and Traum can't be denied, without research surrounding the hardware design history and ethos of these instruments, new developers lack an intellectual framework to build upon in future designs. For this reason, I will seek to collect all available information surrounding Steiner and Traum's wind controller developments since the 1970's to analyze their enduring design influence. More specifically, I seek to find design elements of modern-day wind controllers that are causing them to underperform, as well as the elements that truly innovated the instrument's design.

In order to frame this historical analysis, I will draw on the concepts of technological momentum formed by Thomas P. Hughes (Hughes, 2008), combining the theories of technological determinism & social construction into a more nuanced theory where the sociotechnological relationship has an inertial quality: society tends to drive the design of the technology earlier on, but once the technology obtains momentum in the society, it starts to become more deterministic and influence both itself and society. This concept helps to distinguish which elements are impacted by pre-existing systems (where the sociotechnical system has settled improperly) and can direct the next needed development to revolutionize wind controller production.

To support my argument, I am planning on reaching out to Traum to collect and archive all available design documentation from both him and Steiner. I am also planning on reaching out to the Acoustical Society of America (ASA—of which I am a member and recent meeting attendee) to see if there is any more available information regarding Steiner's abstract and presentation (Musical Acoustics Technical Committee Open Meeting, 2026). I am also hoping that I will then be able to present on this research at the December 2026 ASA meeting in Baltimore.

Conclusion

The deliverable for the technical project will be a fully open-source wind controller design, which will be supported by the STS research paper documenting historical wind controller designs. The goal of both projects is to continue the work of the inventors who have come before me by honoring the genius of their wind controller designs, while also making these design concepts more accessible in the modern maker world. The hope is that this foundational body of research can be continued by both academics and hobbyists to build up the next generation of musicians and engineers. Music has many practical benefits in the scope of engineering education (especially signal processing), but the value of making music as a pursuit of artistry and fun should be held paramount, for the sake of humanity's soul.

Word Count: 1739

References

Berglund Instruments. (2026, May 30). *NuEVI R2*. Retrieved from berglundinstruments.com: https://berglundinstruments.com/nuevi-trumpet-horn-tuba-fingering-instrument/

Bob Kammauff, k. (2026, May 7). *breathcontroller*. Retrieved from github.com: https://github.com/kmw3kx/breathController

Computer Music Specials. (2020, July 7). *The beginner's guide to: MIDI controllers*. Retrieved from MusicRadar: https://www.musicradar.com/tuition/tech/the-beginners-guide-to-midi-controllers-179018

Computer Music Specials. (2020, July 7). *The beginner's guide to: MIDI controllers*. Retrieved from musicradar.com: https://www.musicradar.com/tuition/tech/the-beginners-guide-to-midi-controllers-179018

Dahl, L. (2026). *Sound Synthesis and Control Course Syllabus.* Charlottesville: Department of Music, University of Virginia.

Electrosmith. (2024). *Software - Daisy*. Retrieved from daisy.audio: https://daisy.audio/software/

EWI Series. (2026). Retrieved from akaipro.com: https://www.akaipro.com/products/controllers/electronic-wind/

Hughes, T. P. (2008). Technology and Society. In D. Johnson, & J. Wetmore, *Technology and Society: Building Our Sociotechnical Future* (pp. 141-150). The MIT Press.

Lindgren, B. (2025). *Introduction to Digital Sound Synthesis Course Syllabus.* Charlottesville: Department of Music, University of Virginia.

Musical Acoustics Technical Committee Open Meeting. (2026). *190th Meeting of the Acoustical Society of America.* Philidelphia, PA: Acoustical Society of America.

Steiner, M. (2026). Retrieved from youtube.com: https://www.youtube.com/@marksteiner3493

Steiner, N. (2010, May 30). Nyle Steiner - Interview 2010 / Bach Toccata | Youtube. (J. Hardman, Interviewer)

Steiner, N. A. (2004). The Electronic Valve Instrument (EVI), an electronic musical wind controller for playing synthesizers. *J. Acoust. Soc. Am.*, 115.

Sweetwater.com. (2026, May 30). *Akai Professional EWI Solo Standalone EWI with Built in Speaker*. Retrieved from sweetwater.com: https://www.sweetwater.com/store/detail/EWISolo--akai-professional-ewi-solo-standalone-ewi-with-built-in-speaker

Taki, D. (2020). *A General Method for the Creation of New Electronic Musical Instruments and A New Electronic Wind Instrument.* Urbana, IL : University of Illinois Grainger College of Engineering.

The MIDI Manufacturers Association. (2025). *Spec Detail*. Retrieved from midi.org.

Traum, M. (2025, August 6). *The Nyle Steiner Homepage*. Retrieved from PatchmanMusic.com: https://patchmanmusic.com/NyleSteinerHomepage.html

Traum, M. (2026, April 4). *The Wind Controller FAQ*. Retrieved from PatchmanMusic.com: https://www.patchmanmusic.com/WindControllerFAQ.html

Traum, M. (n.d.). *Nyle Steiner*. Retrieved from Stickband: https://stickbandsj.com/nyle-steiner/